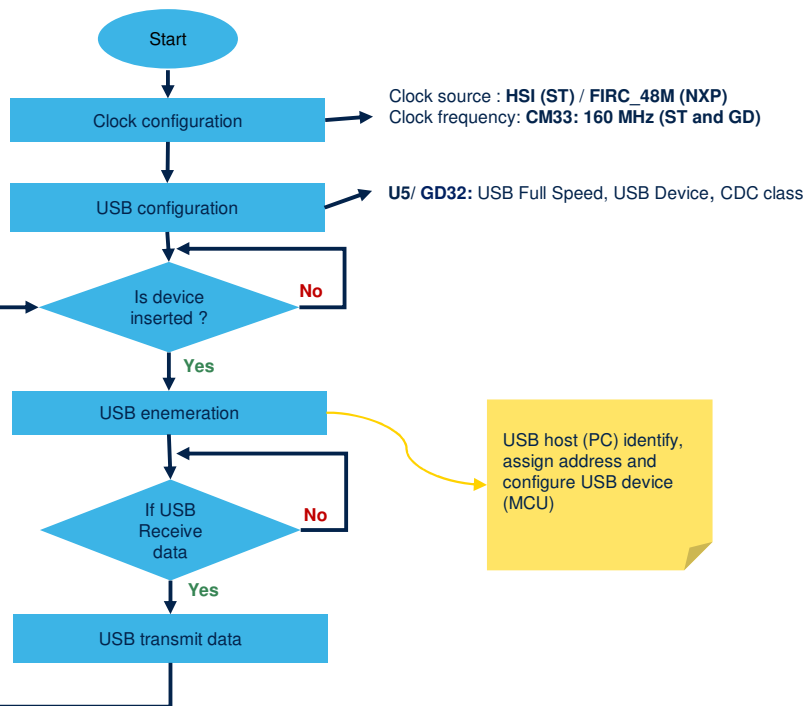


USB Stack (HID,CDC)

STM32 vs GD32 MW Benchmark: USB stack

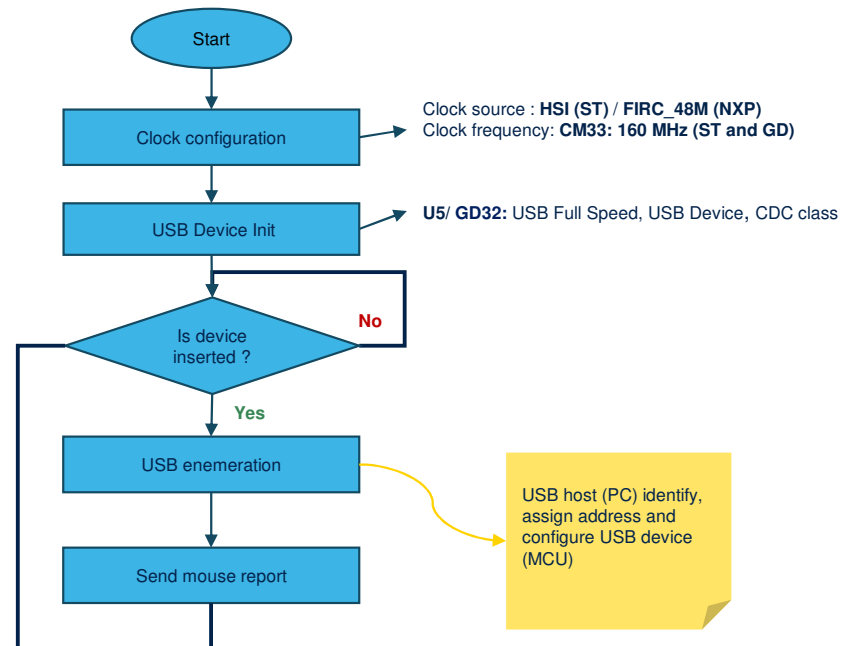
Flowcharts

BareMetal mode : ST/GD USB Library CDC device class STM32U575xx (CM33) vs GD32E507(CM33)



- **USB CDC device class:** USB communication device class allows devices to communicate with computers as serial ports or network interfaces.

BareMetal mode : ST/GD USB Library USB HID device class STM32U575xx (CM33) vs GD32E507(CM33)



- **USB HID device class:** USB Human Interface Device allows to enable communication between device (MCU) and host (PC).

STM32 vs GD32 MW Benchmark: USB stack

Test Results

STM32/GD32 USB Library CDC device (FS)

Footprint (Bytes)

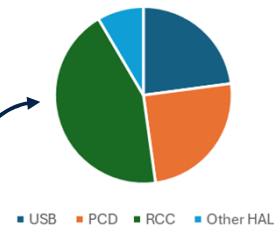
CDC Test Footprint	STM32U575	GD32E507	STM232 vs GD32
ROM	17,049	9,085	+ 88 %
RAM	4,089	9,683	- 57.7 %

ROM Footprint (Bytes)

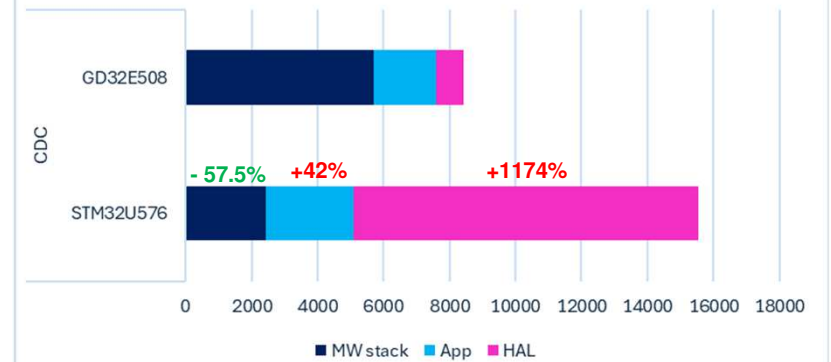
ROM	STM32U575	GD32E507	STM232 vs GD32
App	2,683	1,882	+ 42.5%
HAL	10,446	820	+ 1174%
MW stack	2,423	5,709	- 57.5 %

- STM32 ROM footprint is **higher by average 88%** than GD32
- This is due to:
 - **Application ROM:** STM32 higher by 42% than GD32 because of USB Device user files: usb_device.c, usbd_cdc_if.c, usbd_desc.c, usbd_conf.c (1164 bytes: 43% of total App ROM footprint). In GD32, USB device configuration is managed directly in MW stack
 - **More USB Device configuration flexibility in STM32**
 - **HAL ROM:** STM32 higher by 1174% than GD32, mainly due to:
 - HAL RCC: 4560 bytes => 44% of total HAL ROM footprint
 - HAL/LL USB: 5006 bytes => 48% of total HAL ROM footprint
- The high STM32 App and HAL ROM footprint lead to a lower STM32 USB library ROM footprint than GD32 USB library by 58%

STM32 HAL ROM



USB CDC test ROM footprint



USB CDC Device operations	STM32U575	GD32E507	STM232 vs GD32
Initialization	9.9936	26.07624	-61.68%
Enumeration	218.2	218.5	-0.14%
Transmit data	0.00988	0.010034	-1.53%
Receive data	0.00986	0.009932	-0.72%

- The execution time of STM32U5 USB initialization phase is **lower by 61.68%** than GD32E5
- The execution time of enumeration, transmission and reception phases are **almost close** between STM32U5 and GD32E5.

$$\text{STM32 vs GD32 (\%)} = (\text{STM32} - \text{GD32}) / \text{GD32}$$

STM32 vs GD32 MW Benchmark: USB stack

Test Results

STM32/GD32 USB Library HID device (FS)

Footprint (Bytes)

HID Test Footprint	STM32U575	GD32E507	STM232 vs GD32
ROM	16,269	9,175	+ 77 %
RAM	3,446	9,608	- 65 %

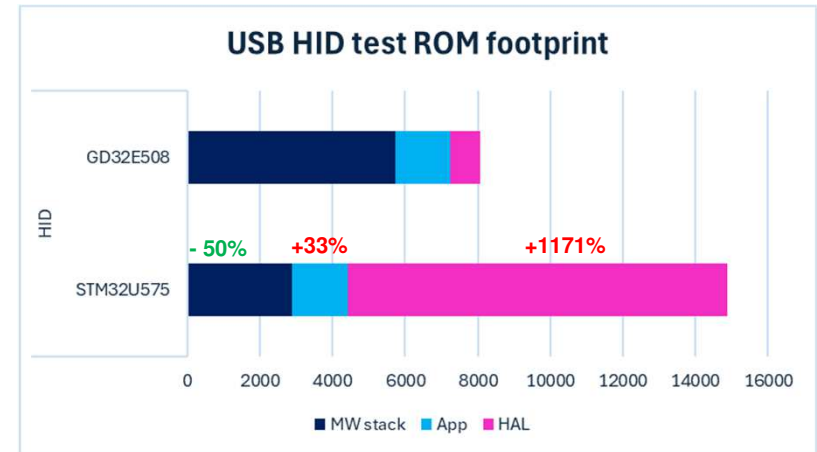
ROM Footprint (Bytes)

ROM	STM32U575	GD32E507	STM232 vs GD32
App	2580	1942	+ 33 %
HAL	10426	820	+ 1171 %
MW stack	2859	5743	- 50 %

- STM32 ROM footprint is **higher by average 77%** than GD32
- This is due to:
 - **Application ROM:** STM32 higher by 33% than GD32 because of USB Device user files: usbd_desc.c, usbd_conf.c, (985 bytes: 38% of total App ROM footprint). In GD32, USB device configuration is managed directly in MW stack
 - **More USB Device configuration flexibility in STM32**
 - **HAL ROM:** STM32 higher by 1171% than GD32, mainly due to:
 - HAL RCC: 4560 bytes => 44% of total HAL ROM footprint
 - HAL/LL USB: 4988 bytes => 48% of total HAL ROM footprint
- The high STM32 App and HAL ROM footprint lead to a lower STM32 USB library ROM footprint than GD32 USB library by 50%

RAM consumption

- RAM footprint analysis shows that the GD32 implementation requires significantly more stack than STM32.
- For both HID and CDC examples, **GD32 needs 0x2000 bytes of stack and 0x2000 bytes of heap**, whereas **STM32 requires only 0x400 bytes of stack and 0x200 bytes of heap** for HID, and the same order of magnitude for CDC.



Execution time (ms)

USB HID Device operations	STM32U575	GD32E507	STM232 vs GD32
Initialization	9.990106	26.08368	-61.70%
Enumeration	213	216	-1.39%
Data transmission	0.009284	0.00935	-0.71%

- The execution time of STM32U5 USB initialization phase is **lower by 61.70 %** than GD32E5
- The execution time of enumeration and data transmission phases are **almost close** between STM32U5 and GD32E5.

$$STM32 \text{ vs } GD32 (\%) = (STM32 - GD32) / GD32$$